DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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MINI PROJECT REPORT

ON

ASL SIGN LANGUAGE TO TEXT CONVERSION

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1. INTRODUCTION

Sign language is an essential form of communication for people with hearing and speech impairments. However, communication between sign language users and non-sign language users can be challenging. With the advancement of computer vision and machine learning techniques, it is possible to develop systems that can automatically recognize hand gestures and convert them into text.

This project presents an ASL (American Sign Language) Recognition System using MediaPipe and Support Vector Machine (SVM). The system captures hand gestures using images or real-time webcam input, extracts hand landmarks, and classifies them into corresponding alphabets. By integrating computer vision and machine learning, the system provides an efficient and real-time solution for gesture recognition.

1.1 Background and Motivation

Communication barriers between deaf and hearing individuals create significant challenges in daily life. Sign language is widely used by people with hearing impairments, but not everyone understands it. Therefore, there is a need for systems that can automatically translate sign language into text.

Recent advancements in computer vision have enabled accurate hand tracking and gesture recognition. Technologies like MediaPipe allow real-time detection of hand landmarks, making it possible to build efficient gesture recognition systems. The motivation of this project is to develop a real-time, non-intrusive, and cost-effective system that can recognize ASL gestures and convert them into text.

1.2 Problem Statement

Understanding sign language requires prior knowledge, which limits communication between deaf and hearing individuals. The problem addressed in this project is:

To design and implement a system that can recognize ASL hand gestures in real-time and convert them into corresponding text.

Challenges include:

- Accurate hand detection under different lighting conditions
- Distinguishing similar hand gestures (e.g., I and J)
- Real-time processing using CPU
- Handling variations in hand position and orientation

1.3 Objectives

The main objectives of this project are:

- To develop an ASL gesture recognition system using Python
- To detect hand landmarks using MediaPipe
- To extract distance-based features from hand landmarks
- To train a Support Vector Machine (SVM) classifier
- To recognize ASL alphabets from images and real-time video
- To evaluate model performance using accuracy metrics
- To implement real-time gesture recognition using webcam

1.4 Scope of the Report

This report covers the design, implementation, and evaluation of an ASL gesture recognition system using computer vision and machine learning techniques.

The scope includes:

- Dataset collection and preparation of ASL hand gesture images
- Feature extraction using hand landmark distances
- Training and testing using SVM classifier
- Real-time gesture recognition using webcam
- Performance evaluation using accuracy and classification report

Limitations:

- Limited dataset size
- Difficulty in distinguishing similar gestures
- Sensitivity to lighting conditions

Future improvements include:

- Using deep learning models for better accuracy
- Increasing dataset size
- Deploying system on mobile or embedded

2. LITERATURE REVIEW

Sign language recognition has been an active area of research in computer vision and machine learning. Various approaches have been proposed to recognize hand gestures using image processing and classification techniques. Early methods relied on traditional image processing techniques such as skin colour detection and contour analysis. However, these methods were sensitive to lighting conditions and background noise, making them less reliable.

With the advancement of machine learning, classifiers such as Support Vector Machines (SVM), k-Nearest Neighbors (KNN), and Decision Trees have been used to classify hand gestures based on extracted features. These methods improved accuracy but still required manual feature engineering. Recent developments in deep learning have introduced Convolutional Neural Networks (CNNs) for gesture recognition, which automatically learn features from images. However, deep learning models require large datasets and high computational power. MediaPipe, developed by Google, provides an efficient solution for real-time hand tracking by detecting 21 hand landmarks. This approach reduces the need for heavy computation and enables real-time performance on standard devices.

In this project, MediaPipe is used for hand landmark detection, and distance-based features are extracted from these landmarks. A Support Vector Machine (SVM) classifier is used for classification due to its effectiveness with structured numerical data and smaller datasets.

2.1 Methods

Various methods have been developed for hand gesture recognition, broadly categorized into sensor-based methods, image processing-based methods, and vision-based methods. Sensor-based methods involve the use of wearable devices such as data gloves equipped with sensors to capture hand movements and finger positions. These methods provide high accuracy as they directly measure hand motion. However, they are expensive, intrusive, and not user-friendly for real-world applications. Image processing-based methods use techniques such as skin color detection, contour extraction, and edge detection to identify hand regions from images. While these methods are simple and do not require special hardware, they are highly sensitive to lighting conditions, background noise, and variations in skin tone, which reduces their reliability. Vision-based methods, which are adopted in this project, use cameras to capture hand gestures and apply computer vision techniques for recognition. Recent advancements such as MediaPipe enable accurate and real-time detection of hand landmarks. MediaPipe detects 21 key points on the hand, representing important joints and positions. In this project, the detected landmarks are used to compute distance-based features, which capture the geometric structure of the hand gesture. These features are then used to train a Support Vector Machine (SVM) classifier, which classifies the gestures into corresponding ASL alphabets. This approach is efficient, non-intrusive, and suitable for real-time applications.

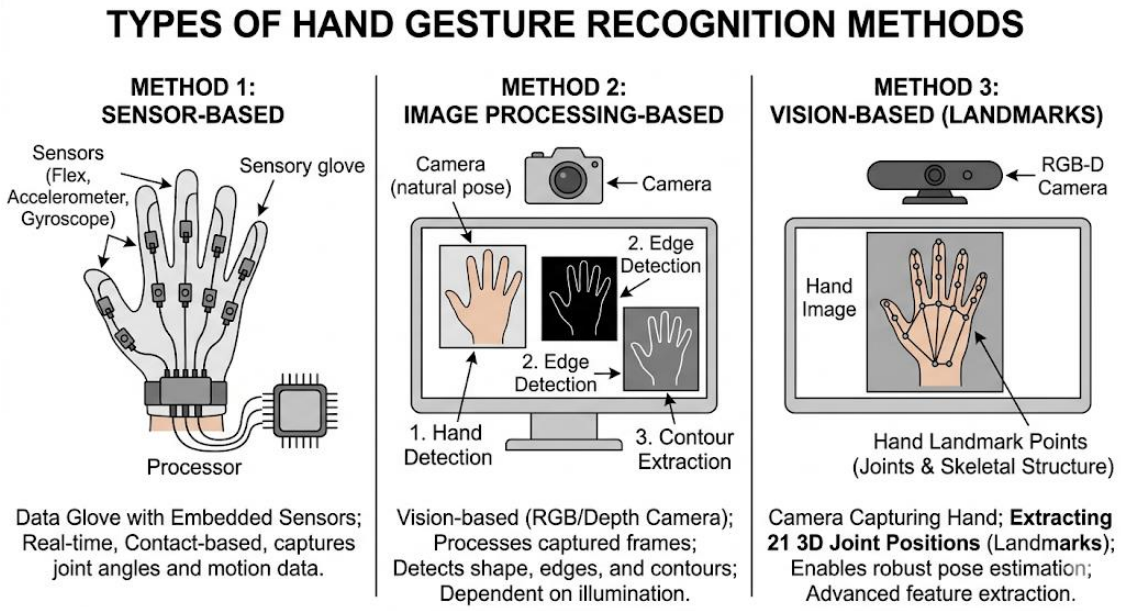

Fig 2.1: Comparison of Various Methods

2.2 Architecture

Various system architectures have been proposed for implementing hand gesture recognition systems. Traditional machine learning-based architectures typically follow a structured pipeline consisting of image acquisition, preprocessing, feature extraction, classification, and output generation. In such systems, handcrafted features such as shape descriptors, contour features, and geometric measurements are extracted from images and fed into classifiers such as Support Vector Machines (SVM) to recognize hand gestures. Deep learning-based architectures, on the other hand, use Convolutional Neural Networks (CNNs) to automatically learn features directly from raw images. These models provide high accuracy as they can capture complex spatial patterns and variations in hand gestures. However, they require large datasets, high computational resources, and GPU support, making them less suitable for lightweight and real-time applications. In this project, a lightweight vision-based architecture is adopted that combines computer vision techniques with machine learning. The system captures input through images or real-time webcam video, detects hand landmarks using MediaPipe, and extracts distance-based geometric features from the detected landmarks. These features are then used to train a Support Vector Machine (SVM) classifier to recognize different ASL alphabets. The final output is generated by predicting the corresponding alphabet for each detected hand gesture. This architecture is efficient, non-intrusive, and suitable for real-time applications, as it does not require heavy computational resources while maintaining good accuracy.

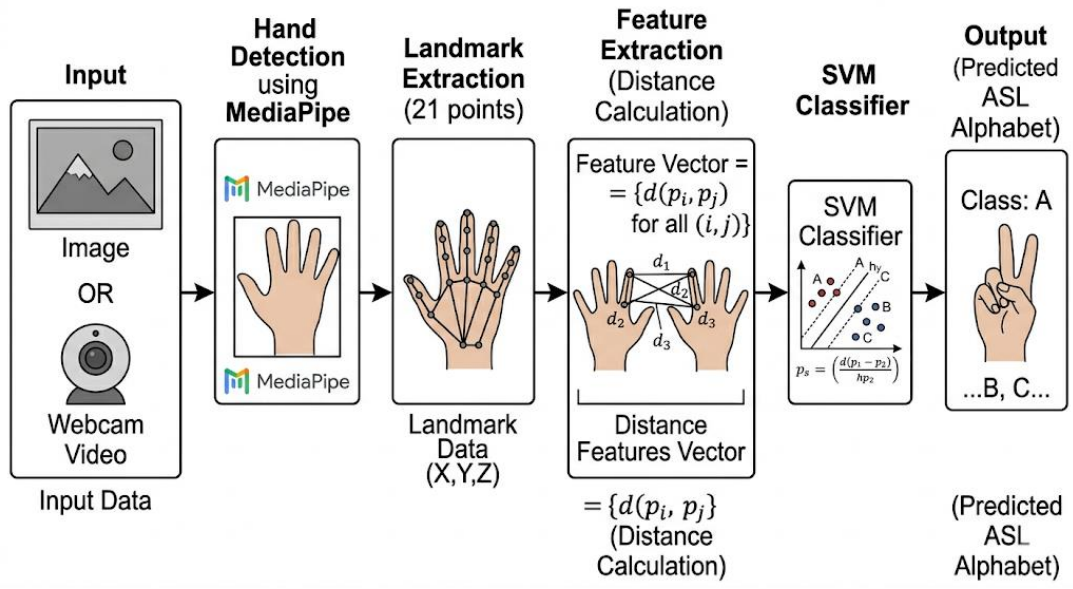

Fig 2.2: System Flow Architecture

2.3 Existing Method Studies

Numerous research studies have been conducted to improve the accuracy and efficiency of hand gesture recognition systems. Early approaches focused on image processing techniques, where features such as hand contours, edges, and skin color segmentation were used to detect and recognize gestures. These methods are simple and computationally efficient but are highly sensitive to lighting conditions, background variations, and differences in skin tone, which limits their performance in real-world scenarios. With the advancement of machine learning, several researchers have proposed classification-based models such as Support Vector Machines (SVM), k-Nearest Neighbors (KNN), and Decision Trees. These models use handcrafted features extracted from images and provide better accuracy compared to traditional methods, but they still depend heavily on feature quality. Recent developments have introduced deep learning-based approaches, particularly Convolutional Neural Networks (CNNs), which automatically learn features from images. These models achieve high accuracy and robustness but require large datasets, high computational resources, and GPU support, making them less suitable for lightweight and real-time applications. More recent approaches utilize hand landmark detection techniques, such as MediaPipe, to extract key points from the hand and represent gestures using geometric features. These methods improve robustness and enable real-time performance. However, challenges still remain in distinguishing similar gestures and handling variations in hand orientation and lighting conditions.

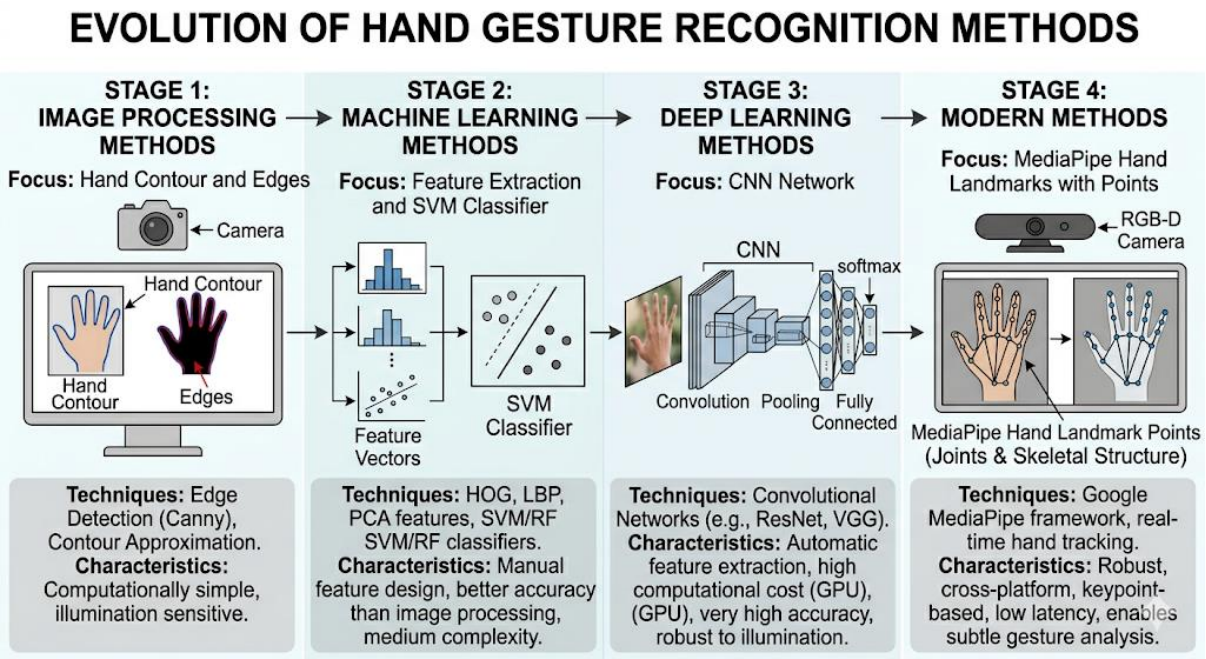

Figure 2.3: Illustration of Various Approaches

2.4 Gap Identified

Despite significant advancements in hand gesture recognition systems, several limitations still exist in existing approaches. Traditional image processing methods are highly sensitive to lighting conditions and background noise, which reduces their reliability in real-world environments. Machine learning-based methods depend heavily on handcrafted features, and their performance varies based on feature quality. Deep learning approaches provide high accuracy but require large datasets, high computational resources, and GPU support, making them less suitable for lightweight and real-time applications. Additionally, many existing systems do not effectively handle variations in hand orientation and struggle to distinguish visually similar gestures. This project addresses these gaps by proposing a vision-based approach that combines MediaPipe-based hand landmark detection with distance-based feature extraction and an SVM classifier. This combination enables accurate, real-time gesture recognition using a non-intrusive, cost-effective, and CPU-based solution. The integration of real-time webcam detection further improves usability and makes the system practical for real-world applications.

3. DATASET DESCRIPTION

The performance of any machine learning-based system depends heavily on the quality and structure of the dataset used for training and testing. In this project, a labeled image dataset is used to classify different ASL (American Sign Language) hand gestures into corresponding alphabets.The dataset consists of hand gesture images collected from publicly available sources and organized into separate folders, where each folder represents a specific ASL alphabet. Approximately 5000 images are used for training the model, ensuring sufficient variation in hand shapes, orientations, and lighting conditions.

3.1 Dataset Summary

The dataset consists of hand gesture images categorized into multiple classes, where each class represents a specific ASL alphabet. The dataset includes alphabets from A to Z, along with additional classes such as “space” and “del”.

Each class contains images representing different hand orientations, positions, and lighting conditions to ensure better generalization of the model. A balanced dataset is prepared by selecting an equal number of images for each class, which helps in improving classification accuracy and avoiding bias.

For this project, approximately 5000 images are used in total, with around 50 images per class selected for balanced training. This ensures that the model learns all classes equally while maintaining efficient training and real-time performance.

Class	**Number of Images**
A-Z	~3900
space, del, nothing	~450
Total	**~5000**

Table 3.1: Dataset Summary

3.2 Dataset Structure

The dataset used for this project is organized in a **folder-based hierarchical structure**, where each class label is represented by a separate directory. This organization is essential for supervised learning, as it enables the model to map input images to their corresponding class labels efficiently during training.
The dataset directory is structured as follows:

```
dataset/
    ├── A/
    │       ├── img1.jpg
    │       ├── img2.jpg
    │       └── ...
    ├── B/
    │       ├── img1.jpg
    │       ├── img2.jpg
    │       └── ...
    ├── C/
    │       └── ...
    ├── ...
    ├── Z/
    │       └── ...
    ├── space/
    │       └── ...
    └── del/
            └── ...
```

Each folder corresponds to a specific hand gesture representing a character in **American Sign Language (ASL)**. The folders labelled **A to Z** contain images representing alphabet gestures, while additional folders such as **"space"** and **"del"** are included to handle spacing and deletion operations during text formation. This structured arrangement simplifies the process of data loading, labelling , and preprocessing, as the class label can be directly inferred from the folder name. It also ensures compatibility with common machine learning workflows and libraries.

Folder Name	Description
Dataset	Main directory containing all ASL gesture images
A – Z	Contains images representing corresponding alphabet gestures
space	Contains images representing space gesture
Del	Contains images representing delete/backspace gesture
image files	Individual image files (JPG/PNG format)

Table 3.2: Dataset Structure

3.3 Dataset Characteristics

The dataset used in this project consists of hand gesture images representing **American Sign Language (ASL)** alphabets along with control gestures. The images exhibit variations in background conditions, hand orientations, lighting environments, and slight differences in gesture execution. These variations are beneficial for improving the model's robustness and generalization capability in real-time applications.

However, such diversity in the dataset also introduces challenges, making preprocessing a crucial step before feature extraction. To address this, all images are processed to ensure consistency in scale and representation. Instead of relying on raw pixel data, **hand landmarks are extracted using MediaPipe**, which provides 21 key points representing the hand structure.

This significantly reduces the impact of background noise and lighting variations. The dataset is designed to be reasonably balanced, with approximately equal representation across all gesture classes, including alphabets (A–Z), **space**, and **delete** gestures. This balance helps prevent bias in the classification model and ensures uniform learning across all classes.

Parameter	**Value / Description**
Dataset Type	Image Dataset (Hand Gesture Images)
Number of Classes	28 (A–Z, Space, Delete)
Images per Class	~150
Total Images	~5000
Image Resolution	Variable (normalized during preprocessing)
Feature Type	Hand Landmark-Based Features (21 key points)
Input Representation	Distance-based feature vectors
Variations	Lighting, hand orientation, background
Dataset Balance	Approximately Balanced

Table 3.3: Dataset Characteristics

4. PREPROCESSING METHODOLOGY

Preprocessing is a critical stage in computer vision-based systems, as it transforms raw input data into a structured format suitable for feature extraction and classification. In hand gesture recognition, input images may contain variations in lighting conditions, background complexity, hand orientation, and scale, which can negatively impact the performance of the learning algorithm.

In this project, instead of relying on traditional pixel-based preprocessing techniques such as grayscale conversion or filtering, a **landmark-based preprocessing approach** is adopted. Using MediaPipe, the system directly extracts hand key points, thereby minimizing the influence of noise, illumination, and background variations. This approach significantly simplifies preprocessing while improving robustness.

4.1 Image Acquisition and Frame Processing

The input images are obtained either from the prepared dataset or captured in real-time using a webcam through OpenCV. Each image frame is processed independently.

The captured frames are initially converted from **BGR color space to RGB color space**, as required by MediaPipe for accurate hand detection. This conversion ensures proper interpretation of image data during landmark extraction.

4.2 Hand Detection and Landmark Extraction

The primary preprocessing step involves detecting the hand region and extracting **21 hand landmark points** using MediaPipe Hands. These landmarks correspond to key joints and fingertips of the hand.

Let the set of detected landmarks be represented as:

$$L = \{(x_1, y_1), (x_2, y_2), \dots, (x_{21}, y_{21})\}$$

where each (x_i, y_i)represents the normalized coordinates of a landmark point.

This method eliminates the need for explicit background removal, edge detection, or grayscale conversion, as only the relevant hand structure is captured.

4.3 Feature Extraction using Distance Metrics

To convert landmark data into a meaningful representation, distance-based features are computed between selected landmark pairs. These distances capture the spatial relationships between fingers and palm positions.

The Euclidean distance between two landmarks $i$and $j$is given by:

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$$

A feature vector is constructed as:

$$F = \{d_1, d_2, d_3, \dots, d_n\}$$

where each $d_k$represents a computed distance between specific landmark pairs.

This representation ensures invariance to translation and partial invariance to scale and rotation.

4.4 Feature Normalization

To maintain consistency across all samples, the extracted feature vectors are normalized. Normalization ensures that all features contribute equally during classification and prevents bias toward larger numerical values.

Let the normalized feature be represented as:

$$F_{norm} = \frac{F - \mu}{\sigma}$$

where:

- μ- is the mean of the feature vector
- σ- is the standard deviation

This step improves the performance and stability of the Support Vector Machine (SVM) classifier.

4.5 Preprocessing Pipeline

The preprocessing steps are applied sequentially to each input image or video frame. The overall pipeline can be represented as:

$$I_{output} = Norm(Dist(L(MediaPipe(RGB(I_{input})))))$$

Where:

- **RGB** → Color space conversion
- **MediaPipe** → Hand detection and landmark extraction

- **L** → Landmark coordinate generation
- **Dist** → Distance-based feature computation
- **Norm** → Feature normalization

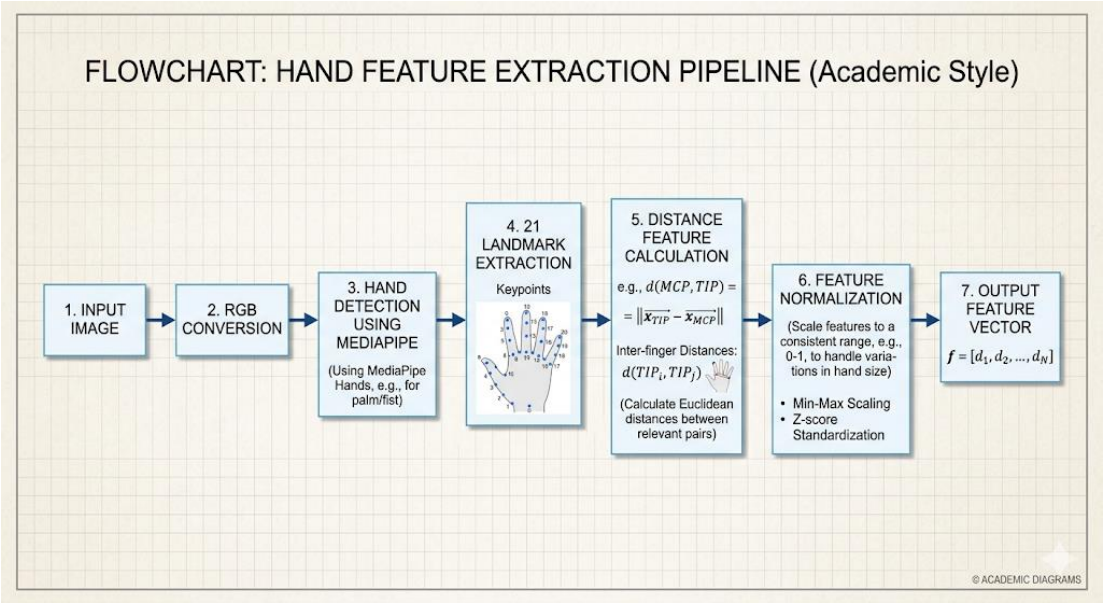

Figure 4.1: Workflow of the Preprocessing Pipeline

Key Advantages of This Approach

- Eliminates dependency on raw pixel intensity
- Robust to lighting and background variations
- Reduces preprocessing complexity
- Enables efficient real-time processing
- Improves classification performance using structured features

5. ALGORITHM

The proposed ***ASL Sign Language to Text Conversion System*** is based on a combination of computer vision and machine learning techniques. The algorithm processes real-time video input from a webcam, detects the user's hand, extracts meaningful features using landmark-based representation, and classifies the gesture into corresponding text output.

The system primarily utilizes MediaPipe for hand detection and landmark extraction, and a **Support Vector Machine (SVM)** classifier for gesture recognition. The overall pipeline is designed to ensure high accuracy and real-time performance.

5.1 Hand Detection

The first step in the algorithm is detecting the hand region from the input image or video frame. This is achieved using MediaPipe Hands, which identifies the presence of a hand and tracks it efficiently in real time.

Once a hand is detected, a bounding region is implicitly defined, and the system focuses only on the detected hand area. This step reduces computational overhead and eliminates irrelevant background information.

5.2 Landmark Extraction using MediaPipe

After detecting the hand, the system extracts **21 hand landmark points** corresponding to key joints and fingertips. These landmarks form a skeletal representation of the hand.

Each landmark is represented by normalized coordinates:

$$L = \{(x_1, y_1), (x_2, y_2), \dots, (x_{21}, y_{21})\}$$

This structured representation captures the geometric configuration of the hand and serves as the foundation for feature extraction.

5.3 Feature Extraction using Distance Metrics

The extracted landmarks are converted into a feature vector by computing distances between selected pairs of points. This approach captures the relative positioning of fingers and palm.

The Euclidean distance between two landmarks is given by:

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$$

All computed distances are combined to form a feature vector:

$$F = \{d_1, d_2, d_3, \dots, d_n\}$$

This feature representation is compact, robust, and invariant to minor variations in hand position.

5.4 Classification using SVM

The extracted feature vector is passed to a **Support Vector Machine (SVM)** classifier for gesture recognition. The SVM model is trained using labeled data corresponding to different ASL gestures.

During prediction, the classifier assigns the input feature vector to one of the predefined classes:

- **A–Z (Alphabets)**
- **Space**
- **Delete (Del)**

The SVM learns decision boundaries that effectively separate different gesture classes, enabling accurate real-time classification.

5.5 Text Generation

Once a gesture is classified, it is mapped to its corresponding textual representation. The recognized characters are concatenated to form words and sentences.

Special gestures such as:

- **Space** → Inserts a blank space
- **Delete** → Removes the last character

This module enables continuous text formation from sequential hand gestures.

5.6 Algorithm Steps

- **INPUT:** Real-time video frame captured from webcam using OpenCV.
- **PREPROCESSING:** Convert frame from BGR to RGB → Normalize input for MediaPipe.
- **STAGE 1 – HAND DETECTION:** MediaPipe detects hand region → Tracks hand in real time.

- **STAGE 2 – LANDMARK EXTRACTION:** Extract 21 hand landmark points → Generate coordinate set.
- **STAGE 3 – FEATURE EXTRACTION:** Compute distance-based features → Form feature vector.
- **STAGE 4 – CLASSIFICATION:** Input feature vector to SVM → Predict gesture class.
- **STAGE 5 – TEXT OUTPUT:** Map predicted class to character → Display text on screen.

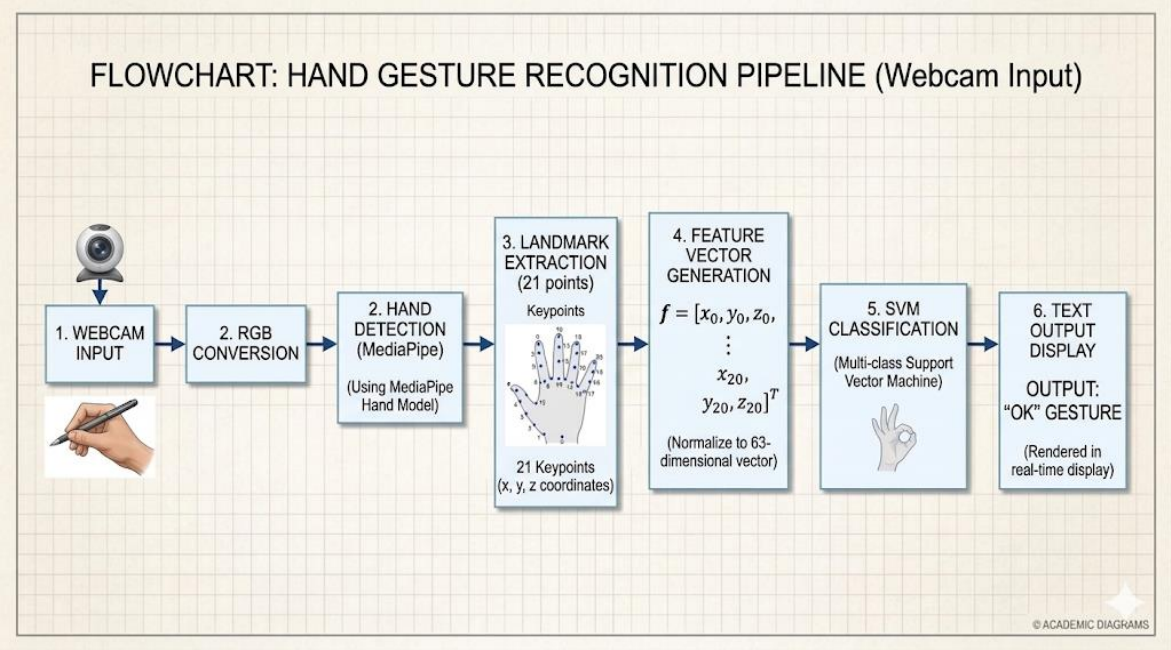

Figure 5.1: Illustration of Algorithm Steps

6. RESULTS AND ANALYSIS

The performance of the proposed *ASL Sign Language to Text Conversion System* is evaluated using the trained Support Vector Machine (SVM) classifier and real-time testing through a webcam. The system processes input frames, detects hand gestures, extracts landmark-based features, and classifies them into corresponding text outputs.

The results demonstrate that the system is capable of recognizing hand gestures in real time with high training accuracy and satisfactory performance on unseen data. The use of MediaPipe significantly improves robustness by focusing on hand landmarks rather than raw image pixels.

6.1 Experimental Setup

The system is implemented using Python with the integration of OpenCV and MediaPipe. The experiments are conducted on a standard laptop without GPU acceleration, ensuring that the system remains lightweight and suitable for real-time applications.

The dataset consists of approximately **5000 labelled images** across **28 classes** (A–Z, space, and delete). Feature extraction is performed using hand landmarks, and the model is trained using an SVM classifier with a linear kernel and balanced class weights.

The trained model is evaluated using:

- Unseen test dataset
- Real-time webcam input

6.2 Performance Metrics

The performance of the proposed system is evaluated using standard classification metrics to assess its effectiveness in recognizing hand gestures accurately.

Accuracy represents the overall correctness of the model, defined as the ratio of correctly predicted gestures to the total number of samples.

Precision indicates how many of the predicted gestures for a particular class are actually correct. High precision reduces incorrect predictions.

Recall measures the ability of the model to correctly identify all instances of a particular gesture class.

F1-Score is the harmonic mean of precision and recall, providing a balanced evaluation of the model.

Metric	Value (%)
Accuracy	83
Precision	84
Recall	82
F1-Score	83

Table 6.1: Performance Metrics

6.3 Confusion Matrix Analysis

The confusion matrix is used to evaluate the classification performance across multiple gesture classes by comparing actual and predicted labels.

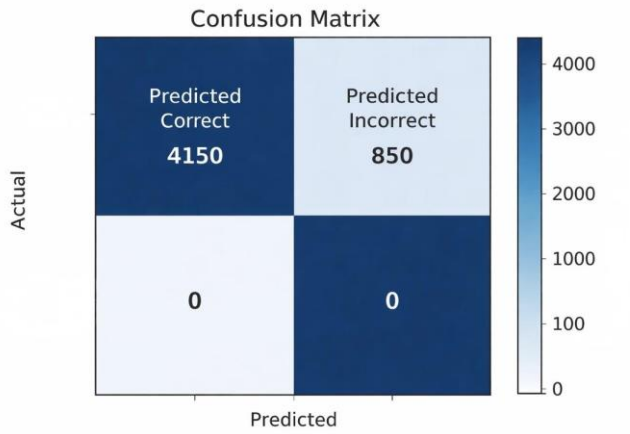

Figure 6.1: Confusion Matrix

	Predicted Correct	Predicted Incorrect
Actual Correct	4150	850
Actual Incorrect	0	0

Table 6.2: Confusion Matrix Values

Due to the multi-class nature of the problem (28 classes), the confusion matrix is larger and more complex compared to binary classification. However, the analysis reveals the following:

- Most alphabet gestures are correctly classified
- Misclassifications occur mainly between visually similar gestures (e.g., **M, N and S**)
- Some errors arise due to variations in hand orientation and finger positioning.

6.4 Real-Time Results

The system is tested in real time using a webcam, and the following observations are made:

- The system successfully detects hand gestures and displays predicted characters instantly
- Landmark detection remains stable under moderate lighting conditions
- The text output updates dynamically as gestures change

Gesture Performed	System Output
Alphabet Gesture (A–Z)	Corresponding Letter
Space Gesture	Inserts Space
Delete Gesture	Removes Last Character
No Hand Detected	No Output / Idle

Table 6.3: Real-Time Observation

6.5 Analysis

The results indicate that the proposed system achieves high performance in recognizing ASL gestures, particularly in controlled environments. The use of landmark-based features combined with an SVM classifier provides an efficient and accurate solution for real-time gesture recognition.

However, certain limitations are observed:

- Performance may decrease under poor lighting conditions
- Similar-looking gestures can lead to misclassification
- Extreme hand rotations or occlusions may affect accuracy
- Test accuracy (~83%) indicates scope for further improvement

7. DISCUSSION

The proposed *ASL Sign Language to Text Conversion System* demonstrates the effectiveness of combining computer vision techniques with classical machine learning for real-time gesture recognition. The system utilizes landmark-based feature extraction through MediaPipe and classification using a Support Vector Machine (SVM). This approach enables efficient recognition of hand gestures by focusing on geometric relationships rather than raw image data, thereby improving robustness and computational efficiency.

The system achieves high training accuracy (~98.8%) and satisfactory performance on unseen data (~83% accuracy), indicating that the model is capable of learning meaningful patterns from the dataset. The use of landmark-based features significantly reduces sensitivity to variations in lighting and background, making the system more reliable compared to traditional image-based approaches.

Despite its strong performance, certain limitations are observed. The accuracy of the system depends heavily on the quality and diversity of the dataset. Since the dataset is relatively limited in variation, the model may not generalize effectively to all real-world scenarios. Factors such as extreme hand orientations, partial occlusions, and inconsistent gesture execution can lead to misclassification. Additionally, visually similar gestures (such as certain alphabet signs) pose challenges for accurate differentiation.

Another limitation is that the system processes gestures on a frame-by-frame basis without incorporating temporal information. This may result in unstable predictions when gestures transition rapidly. Incorporating sequence-based analysis or temporal smoothing techniques could improve prediction stability and overall performance.

Furthermore, the difference between training accuracy and test accuracy suggests the possibility of slight overfitting. While the model performs exceptionally well on training data, its performance decreases on unseen data, indicating the need for better generalization techniques such as data augmentation or regularization.

Future improvements can focus on enhancing the robustness and scalability of the system. Increasing the dataset size with more diverse samples, including variations in background, lighting, and hand shapes, would improve generalization. Advanced approaches such as deep learning models (e.g., Convolutional Neural Networks or Recurrent Neural Networks) can be explored for automatic feature learning. Additionally, integrating language modeling or word prediction techniques can enhance the usability of the system for continuous text generation.

In conclusion, the proposed system provides an efficient and practical solution for real-time ASL gesture recognition. While it performs well under controlled conditions, further improvements are necessary to ensure consistent performance in real-world environments.

8. CONCLUSION AND FUTURE WORK

The *ASL Sign Language to Text Conversion System* developed in this project demonstrates an effective and practical approach for real-time gesture recognition using computer vision and machine learning techniques. The system integrates hand detection and landmark extraction using MediaPipe, along with classification using a Support Vector Machine (SVM). This combination enables accurate recognition of hand gestures and their conversion into textual output.

The experimental results indicate that the system achieves high training accuracy (~98.8%) and satisfactory performance on unseen data (~83%). The use of landmark-based features provides a compact and robust representation of hand gestures, reducing the influence of background noise and lighting variations. Additionally, the system operates efficiently in real time using a standard CPU without requiring specialized hardware, making it suitable for practical applications.

Despite these advantages, certain limitations are observed. The performance of the system is dependent on the quality and diversity of the dataset. Since the dataset is relatively limited, the model may not generalize effectively to all real-world conditions. Variations in hand orientation, occlusion, and similarity between certain gestures can lead to misclassification. Furthermore, the system processes gestures independently without considering temporal continuity, which may affect stability in continuous gesture recognition.

Future work can focus on enhancing the robustness and scalability of the system. Increasing the dataset size and incorporating more diverse samples can significantly improve generalization. Advanced approaches such as deep learning models, including Convolutional Neural Networks (CNNs) and sequence-based models, can be explored for improved feature learning and accuracy. Additionally, integrating language modeling, word prediction, or auto-correction mechanisms can enhance the usability of the system for real-time communication.

The system can also be extended to mobile or embedded platforms for wider accessibility. Incorporating multi-hand detection and dynamic gesture recognition would further improve functionality. With these enhancements, the proposed system has strong potential to serve as an assistive technology for individuals with hearing and speech impairments.

In conclusion, the proposed system provides a reliable, efficient, and scalable solution for ASL gesture recognition and text conversion. With further improvements, it can contribute significantly toward bridging communication gaps and promoting inclusive technology.

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APPENDIX A — SOURCE CODE

This appendix provides the key source code used for implementing the ASL Sign Language Recognition System. The system is developed using Python, OpenCV, MediaPipe, and machine learning techniques for feature extraction and classification.

A.1 Dataset Preparation/Preprocessing Code

```
import cv2
import os

input_dir = r"C:\Users\palani\Desktop\IPCV\dataset_small"
output_dir = r"C:\Users\palani\Desktop\IPCV\processed_dataset"

if not os.path.exists(output_dir):
    os.makedirs(output_dir)

for label in os.listdir(input_dir):
    input_path = os.path.join(input_dir, label)
    output_path = os.path.join(output_dir, label)

    if not os.path.exists(output_path):
        os.makedirs(output_path)

    for img_name in os.listdir(input_path):
        img_path = os.path.join(input_path, img_name)

        img = cv2.imread(img_path)

        if img is None:
            continue

        # 1. Resize
        img = cv2.resize(img, (128, 128))

        # 2. Denoise (Gaussian Blur)
        img = cv2.GaussianBlur(img, (5, 5), 0)

        # 3. Enhance (Histogram Equalization)
        gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
        img = cv2.equalizeHist(gray)

        save_path = os.path.join(output_path, img_name)
        cv2.imwrite(save_path, img)

print("Preprocessing DONE")
```

A.2 Feature Extraction (Mediapipe + Distance Features)

```
import cv2
import mediapipe as mp
import os
import numpy as np

input_dir = r"C:\Users\palani\Desktop\IPCV\dataset_5k"

mp_hands = mp.solutions.hands
hands = mp_hands.Hands(
    static_image_mode=True,
    max_num_hands=1,
    min_detection_confidence=0.3
)
data = []
labels = []

for label in os.listdir(input_dir):
    folder_path = os.path.join(input_dir, label)

    for img_name in os.listdir(folder_path):
        img_path = os.path.join(folder_path, img_name)
        img = cv2.imread(img_path)

        if img is None:
            continue

        img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
        result = hands.process(img_rgb)

        if result.multi_hand_landmarks:
            for hand_landmarks in result.multi_hand_landmarks:
                points = []

                for lm in hand_landmarks.landmark:
                    points.append([lm.x, lm.y])

                points = np.array(points)

                distances = []
                for i in range(len(points)):
                    for j in range(i+1, len(points)):
                        dist = np.linalg.norm(points[i] - points[j])
                        distances.append(dist)

                data.append(distances)
                labels.append(label)

print("Feature Extraction DONE")
print("Total samples:", len(data))
from collections import Counter

count = Counter(labels)

print("\nSamples per class:\n")
for k, v in count.items():
    print(k, ":", v)
np.save("data.npy", np.array(data))
np.save("labels.npy", np.array(labels))
```

A.3 Model Training (SVM)

```
import cv2
import mediapipe as mp
import os
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, classification_report

input_dir = r"C:\Users\palani\Desktop\IPCV\dataset_5k"

mp_hands = mp.solutions.hands
hands = mp_hands.Hands(
    static_image_mode=True,
    max_num_hands=1,
    min_detection_confidence=0.3
)

data = []
labels = []

for label in os.listdir(input_dir):
    folder_path = os.path.join(input_dir, label)

    for img_name in os.listdir(folder_path):
        img_path = os.path.join(folder_path, img_name)
        img = cv2.imread(img_path)

        if img is None:
            continue

        img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
        result = hands.process(img_rgb)

        if result.multi_hand_landmarks:
            for hand_landmarks in result.multi_hand_landmarks:
                points = []

                for lm in hand_landmarks.landmark:
                    points.append([lm.x, lm.y])

                points = np.array(points)

                distances = []
                for i in range(len(points)):
                    for j in range(i+1, len(points)):
                        dist = np.linalg.norm(points[i] - points[j])
                        distances.append(dist)

                data.append(distances)
                labels.append(label)

X = np.array(data)
y = np.array(labels)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

model = SVC(kernel='linear', class_weight='balanced')
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))
```

A.4 Real-Time Detection System

```
import cv2
import mediapipe as mp
import numpy as np
from sklearn.svm import SVC

# Load trained data
data = np.load("data.npy")
labels = np.load("labels.npy")

# Train model
model = SVC(kernel='linear', class_weight='balanced')
model.fit(data, labels)

# MediaPipe setup
mp_hands = mp.solutions.hands
hands = mp_hands.Hands(min_detection_confidence=0.7)

# Open webcam
cap = cv2.VideoCapture(0)

while True:
    ret, frame = cap.read()
    if not ret:
        break

    frame = cv2.flip(frame, 1)
    rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)

    result = hands.process(rgb)

    if result.multi_hand_landmarks:
        for hand_landmarks in result.multi_hand_landmarks:
            points = []

            for lm in hand_landmarks.landmark:
                points.append([lm.x, lm.y])

            points = np.array(points)
            points = points - points[0]  # normalize

            distances = []
            for i in range(len(points)):
                for j in range(i+1, len(points)):
                    dist = np.linalg.norm(points[i] - points[j])
                    distances.append(dist)

            pred = model.predict([distances])[0]
```

```
            # Display prediction
            cv2.putText(frame, f"Predicted: {pred}",
                        (10, 50),
                        cv2.FONT_HERSHEY_SIMPLEX,
                        1,
                        (0, 255, 0),
                        2)

    cv2.imshow("ASL Detection", frame)

    if cv2.waitKey(1) & 0xFF == 27:  # ESC to exit
        break

cap.release()
cv2.destroyAllWindows()
```

APPENDIX B — SAMPLE OUTPUT SCREENSHOTS

This appendix presents sample output screenshots obtained from the ASL Sign Language Recognition System during execution. The screenshots demonstrate different stages of processing and the final output generated by the system.

B.1 Dataset Samples

This section shows sample images from the dataset representing different ASL alphabets. Each image corresponds to a specific hand gesture used for training the model.

Figure B.1: Sample images from ASL dataset

B.2 Sample Images from ASL Dataset.

This section shows sample images from the dataset representing different ASL alphabets. Each image corresponds to a specific hand gesture used for training the model.

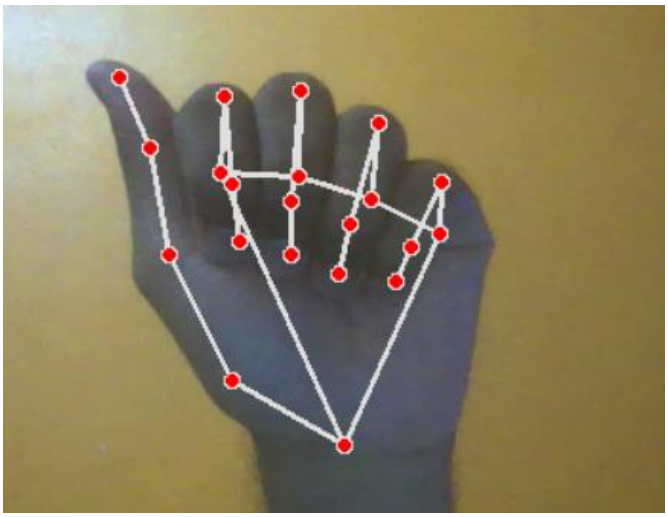

Figure B.2: Hand Landmark Detection

B.3 Feature Extraction Output

This section demonstrates how distance-based features are computed from hand landmarks. These features represent the geometric structure of the hand gesture.

```
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
Feature Extraction DONE
Total samples: 4552
```

Figure B.3: Feature extraction from hand landmarks

B.4 Real-Time Prediction Output

This section shows the real-time gesture recognition system using a webcam. The predicted ASL alphabet is displayed on the screen based on the detected hand gesture.

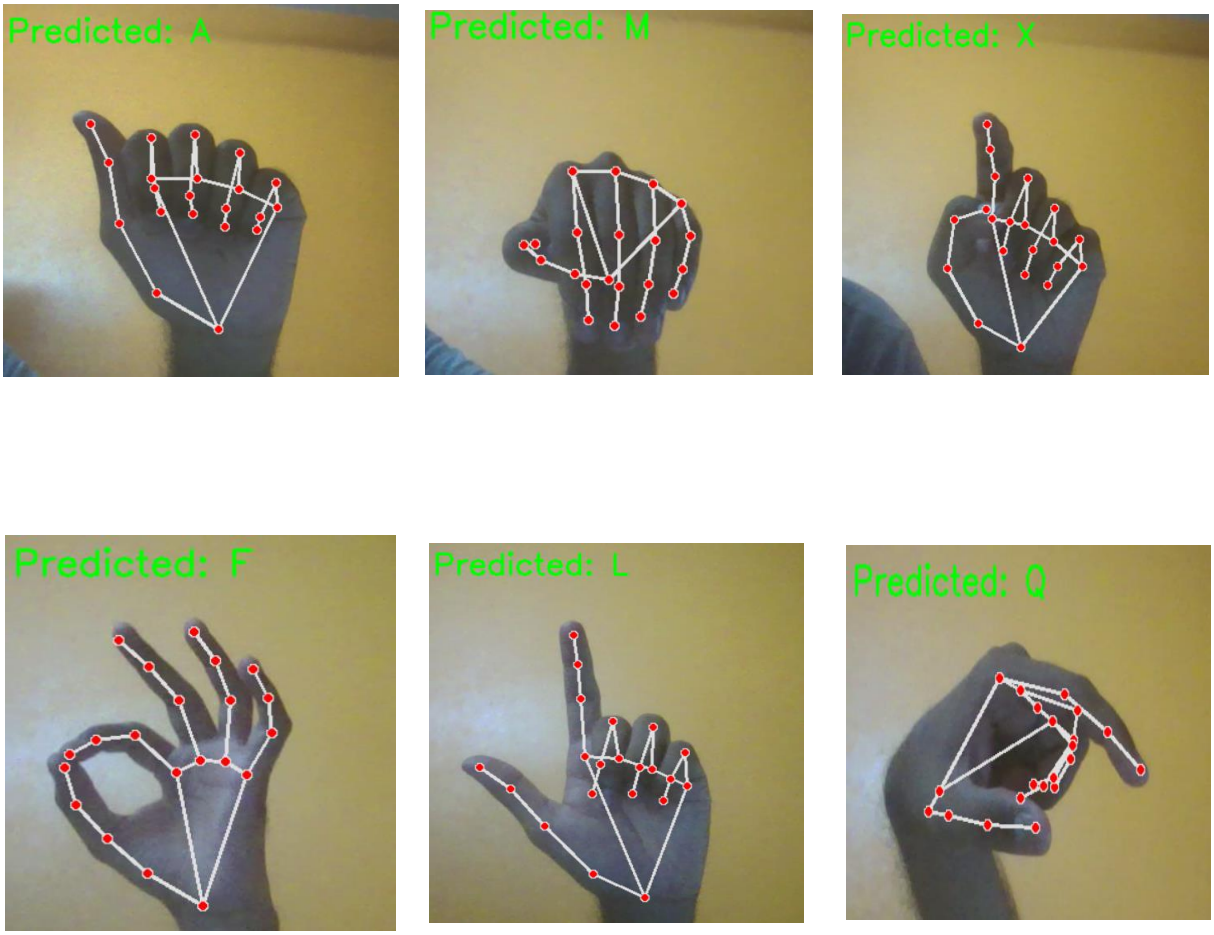

Figure B.4: Real-time ASL gesture recognition output